

Backend Database Schema

DICE by Sanyog — PostgreSQL Schema

Version: 1.0.0

Date: June 2025

ORM: Prisma

Database: PostgreSQL 15

1. Entity Relationship Overview

```
users
├──< applications (userId)
│   ├──< application_documents (applicationId)
│   ├──< application_notes (applicationId)
│   ├──< application_queries (applicationId)
│   ├──< payments (applicationId)
│   └──< technical_reviews (applicationId)
│       ├──< tr_findings (technicalReviewId)
│       └──< tr_queries (technicalReviewId)
├──< certificates (userId)
├──< notifications (userId)
├──< bookmarks (userId)
│   └──> insights (insightId)
└──< push_tokens (userId)
```

insights (standalone, seeded by admin/AI pipeline)

labs (standalone reference table)

certification_bodies (standalone reference table)

2. Table Definitions

2.1 users

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK, default gen_random_uuid()	Primary key
phone	VARCHAR(15)	UNIQUE, NOT NULL	Mobile number (E.164 format)
email	VARCHAR(255)	UNIQUE, nullable	Optional email
name	VARCHAR(100)	NOT NULL	Full name
company_name	VARCHAR(150)	NOT NULL	Business name
role	ENUM	NOT NULL, default 'viewer'	admin , manager , viewer
gst_number	VARCHAR(15)	nullable, UNIQUE	GST registration number
pan_number	VARCHAR(10)	nullable	PAN card number
address	TEXT	nullable	Street address
city	VARCHAR(100)	nullable	City
state	VARCHAR(100)	nullable	State
pincode	VARCHAR(6)	nullable	PIN code
country	VARCHAR(100)	default 'India'	Country
subscription	ENUM	NOT NULL, default 'free'	free , pro , enterprise
business_role	VARCHAR(50)	nullable	manufacturer , importer , exporter , consultant
industries	TEXT[]	default '{}'	Array of industry strings
target_markets	TEXT[]	default '{}'	Array of market strings
cert_interests	TEXT[]	default '{}'	Array: BIS, EPR, WPC, etc.
goals	TEXT[]	default '{}'	Onboarding goals
is_verified	BOOLEAN	default false	Phone verified

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>is_onboarded</code>	<code>BOOLEAN</code>	default <code>false</code>	Completed onboarding
<code>avatar_url</code>	<code>TEXT</code>	nullable	Profile picture URL (S3)
<code>subscription_expires_at</code>	<code>TIMESTAMPTZ</code>	nullable	Subscription expiry
<code>created_at</code>	<code>TIMESTAMPTZ</code>	NOT NULL, default <code>now()</code>	Creation timestamp
<code>updated_at</code>	<code>TIMESTAMPTZ</code>	NOT NULL, default <code>now()</code>	Last update timestamp
<code>deleted_at</code>	<code>TIMESTAMPTZ</code>	nullable	Soft delete timestamp

Indexes:

```
CREATE UNIQUE INDEX idx_users_phone ON users(phone) WHERE deleted_at IS NULL;
CREATE UNIQUE INDEX idx_users_email ON users(email) WHERE email IS NOT NULL AND deleted_at IS NULL;
CREATE INDEX idx_users_subscription ON users(subscription);
CREATE INDEX idx_users_created_at ON users(created_at);
```

2.2 `refresh_tokens`

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	PK	Primary key
<code>user_id</code>	<code>UUID</code>	FK → users.id, NOT NULL	Owner
<code>token_hash</code>	<code>VARCHAR(64)</code>	UNIQUE, NOT NULL	SHA-256 hash of token
<code>expires_at</code>	<code>TIMESTAMPTZ</code>	NOT NULL	Expiry (30 days)
<code>revoked_at</code>	<code>TIMESTAMPTZ</code>	nullable	If manually revoked
<code>device_info</code>	<code>JSONB</code>	nullable	Platform, OS, device model
<code>created_at</code>	<code>TIMESTAMPTZ</code>	default <code>now()</code>	

Indexes:

```
CREATE INDEX idx_refresh_tokens_user_id ON refresh_tokens(user_id);
CREATE INDEX idx_refresh_tokens_expires_at ON refresh_tokens(expires_at);
```

2.3 otp_attempts

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
phone	VARCHAR(15)	NOT NULL	Target phone
otp_hash	VARCHAR(64)	NOT NULL	SHA-256 of OTP
attempts	SMALLINT	default 0	Failed attempts (max 3)
expires_at	TIMESTAMPTZ	NOT NULL	TTL: 5 minutes
verified_at	TIMESTAMPTZ	nullable	When OTP was used
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_otp_phone_expires ON otp_attempts(phone, expires_at);
```

2.4 push_tokens

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, NOT NULL	Owner
token	TEXT	NOT NULL	FCM/APNs device token
platform	ENUM	NOT NULL	ios , android
is_active	BOOLEAN	default true	
last_used_at	TIMESTAMPTZ	nullable	
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE UNIQUE INDEX idx_push_tokens_user_token ON push_tokens(user_id, token);
CREATE INDEX idx_push_tokens_user_active ON push_tokens(user_id) WHERE is_active = true;
```

2.5 certification_bodies

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
name	VARCHAR(200)	NOT NULL	CB name
short_name	VARCHAR(50)	NOT NULL	Abbreviation
logo_url	TEXT	nullable	Logo (S3)
website	TEXT	nullable	Official website
address	TEXT	nullable	Physical address
email	VARCHAR(255)	nullable	Contact email
phone	VARCHAR(15)	nullable	Contact phone
accreditations	TEXT[]	default '{}'	NABL, BIS-approved, etc.
cert_types	TEXT[]	default '{}'	BIS, EPR, WPC, etc.
avg_tat_days	SMALLINT	nullable	Average turnaround days
success_rate	DECIMAL(5,2)	nullable	% success rate
pricing_tier	ENUM	nullable	budget , mid , premium
is_active	BOOLEAN	default true	
is_sanyog_partner	BOOLEAN	default false	Sanyog preferred CB
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

2.6 Labs

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
name	VARCHAR(200)	NOT NULL	Lab name
short_name	VARCHAR(50)	nullable	
nabl_no	VARCHAR(50)	nullable	NABL accreditation number
address	TEXT	nullable	
city	VARCHAR(100)	nullable	
state	VARCHAR(100)	nullable	
pincode	VARCHAR(6)	nullable	
email	VARCHAR(255)	nullable	
phone	VARCHAR(15)	nullable	
test_categories	TEXT[]	default '{}'	Electronics, Food, Chemical, etc.
cert_types	TEXT[]	default '{}'	BIS, WPC, FSSAI, etc.
avg_tat_days	SMALLINT	nullable	
is_active	BOOLEAN	default true	
created_at	TIMESTAMPTZ	default now()	

2.7 certificates

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, NOT NULL	Owner
application_id	UUID	FK → applications.id, nullable	Source application
cert_no	VARCHAR(100)	UNIQUE, NOT NULL	Certificate number
type	ENUM	NOT NULL	BIS , EPR , WPC , FSSAI , ISO , CE
standard	VARCHAR(100)	NOT NULL	IS/IEC/EN standard number
product	VARCHAR(200)	NOT NULL	Product name
product_category	VARCHAR(100)	nullable	Electronics, Food, etc.
model_no	VARCHAR(100)	nullable	Product model
issued_date	DATE	NOT NULL	
expiry_date	DATE	NOT NULL	
issuing_authority	VARCHAR(200)	NOT NULL	BIS HQ / CPCB, etc.
manufacturer	VARCHAR(200)	NOT NULL	
manufacturer_address	TEXT	nullable	
cb_id	UUID	FK → certification_bodies.id, nullable	Associated CB
lab_id	UUID	FK → labs.id, nullable	Testing lab
status	ENUM	NOT NULL, default 'active'	active , inactive , expired , revoked
file_url	TEXT	nullable	Certificate PDF (S3)
qr_code_url	TEXT	nullable	QR verification image (S3)
verification_url	TEXT	nullable	Public verify URL
reminder_sent_30d	BOOLEAN	default false	

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
			30-day expiry reminder sent
reminder_sent_60d	BOOLEAN	default false	60-day expiry reminder sent
reminder_sent_90d	BOOLEAN	default false	90-day expiry reminder sent
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_certificates_user_id ON certificates(user_id);
CREATE INDEX idx_certificates_status ON certificates(status);
CREATE INDEX idx_certificates_expiry ON certificates(expiry_date) WHERE status = 'active';
CREATE INDEX idx_certificates_type ON certificates(type);
```

2.8 applications

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
app_no	VARCHAR(20)	UNIQUE, NOT NULL	Auto-generated: SCS-YYYY-XXXX
user_id	UUID	FK → users.id, NOT NULL	Applicant
cert_type	ENUM	NOT NULL	BIS , EPR , WPC , FSSAI , ISO , CE
product	VARCHAR(200)	NOT NULL	Product name
product_category	VARCHAR(100)	nullable	
standard	VARCHAR(100)	NOT NULL	Target standard/IS number
model_no	VARCHAR(100)	nullable	
hsn_code	VARCHAR(10)	nullable	HSN/SAC code
status	ENUM	NOT NULL, default 'draft'	draft , submitted , under_review , lab_testing , technical_review , approved , rejected , completed
progress	SMALLINT	NOT NULL, default 0	0-100 completion %
lab_id	UUID	FK → labs.id, nullable	Selected test lab
cb_id	UUID	FK → certification_bodies.id, nullable	Selected CB
assigned_to	VARCHAR(100)	nullable	Sanyog consultant name
assigned_officer	VARCHAR(100)	nullable	Government officer
submitted_date	TIMESTAMPZ	nullable	When formally submitted
estimated_completion	DATE	nullable	Expected cert date
amount	DECIMAL(10,2)	NOT NULL, default 0	Total fee

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
paid_amount	DECIMAL(10,2)	NOT NULL, default 0	Amount paid
rejection_reason	TEXT	nullable	If rejected
priority	ENUM	default 'normal'	low , normal , high , urgent
metadata	JSONB	nullable	Flexible additional fields
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	
deleted_at	TIMESTAMPTZ	nullable	Soft delete

Indexes:

```
CREATE UNIQUE INDEX idx_applications_app_no ON applications(app_no);
CREATE INDEX idx_applications_user_id ON applications(user_id);
CREATE INDEX idx_applications_status ON applications(status);
CREATE INDEX idx_applications_cert_type ON applications(cert_type);
CREATE INDEX idx_applications_created_at ON applications(created_at);
```

Sequence for app_no:

```
CREATE SEQUENCE app_seq START 1001;
-- app_no generated as: 'SCS-' || EXTRACT(YEAR FROM NOW()) || '-' || LPAD(nextval('app_seq')::text, 4, '0')
```

2.9 application_timeline

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	
status	VARCHAR(50)	NOT NULL	Status at this point
title	VARCHAR(200)	NOT NULL	Display title
description	TEXT	nullable	Details
actor	VARCHAR(100)	nullable	Who performed the action
event_date	TIMESTAMPTZ	NOT NULL	When this happened
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_timeline_app_id ON application_timeline(application_id);
CREATE INDEX idx_timeline_event_date ON application_timeline(event_date);
```

2.10 application_documents

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	Parent application
user_id	UUID	FK → users.id, NOT NULL	Uploader
name	VARCHAR(200)	NOT NULL	Document name
category	VARCHAR(50)	NOT NULL	identity , product , test_report , lab , govt , other
file_url	TEXT	NOT NULL	S3 URL
file_key	TEXT	NOT NULL	S3 object key
file_size	INTEGER	NOT NULL	Size in bytes
mime_type	VARCHAR(100)	NOT NULL	MIME type
status	ENUM	default 'pending'	pending , approved , rejected
rejection_reason	TEXT	nullable	
reviewed_by	VARCHAR(100)	nullable	
reviewed_at	TIMESTAMPTZ	nullable	
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_app_docs_application_id ON application_documents(application_id);
CREATE INDEX idx_app_docs_user_id ON application_documents(user_id);
```


2.11 application_notes

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	
user_id	UUID	FK → users.id, NOT NULL	Author
text	TEXT	NOT NULL	Note content
is_internal	BOOLEAN	default false	Sanyog-internal note
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_app_notes_application_id ON application_notes(application_id);
```

2.12 application_queries

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	
question	TEXT	NOT NULL	Query text
raised_by	VARCHAR(100)	NOT NULL	Department / officer name
raised_at	TIMESTAMPTZ	NOT NULL	
status	ENUM	default 'open'	open , resolved
response	TEXT	nullable	User's response
responded_by	UUID	FK → users.id, nullable	
responded_at	TIMESTAMPTZ	nullable	
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_app_queries_application_id ON application_queries(application_id);
CREATE INDEX idx_app_queries_status ON application_queries(status);
```

2.13 `technical_reviews`

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	PK	Primary key
<code>tr_no</code>	<code>VARCHAR(30)</code>	UNIQUE, NOT NULL	TR number: TR-YYYY-XXXX
<code>application_id</code>	<code>UUID</code>	FK → applications.id, NOT NULL	
<code>status</code>	<code>ENUM</code>	default <code>'pending'</code>	<code>pending</code> , <code>in_progress</code> , <code>completed</code> , <code>failed</code>
<code>review_date</code>	<code>DATE</code>	nullable	Scheduled/actual review date
<code>reviewed_by</code>	<code>VARCHAR(100)</code>	nullable	BIS/authority reviewer name
<code>assigned_officer</code>	<code>VARCHAR(100)</code>	nullable	
<code>lab_report_no</code>	<code>VARCHAR(100)</code>	nullable	
<code>lab_id</code>	<code>UUID</code>	FK → labs.id, nullable	
<code>summary</code>	<code>TEXT</code>	nullable	Overall review summary
<code>pass_count</code>	<code>SMALLINT</code>	default <code>0</code>	Computed: findings with pass
<code>fail_count</code>	<code>SMALLINT</code>	default <code>0</code>	Computed: findings with fail
<code>pending_count</code>	<code>SMALLINT</code>	default <code>0</code>	Computed: pending findings
<code>created_at</code>	<code>TIMESTAMPZ</code>	default <code>now()</code>	
<code>updated_at</code>	<code>TIMESTAMPZ</code>	default <code>now()</code>	

Indexes:

```
CREATE INDEX idx_tr_application_id ON technical_reviews(application_id);
CREATE INDEX idx_tr_status ON technical_reviews(status);
```

2.14 `tr_findings`

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	PK	Primary key
<code>technical_review_id</code>	<code>UUID</code>	FK → <code>technical_reviews.id</code> , NOT NULL	
<code>clause</code>	<code>VARCHAR(50)</code>	NOT NULL	Clause reference (e.g., IS 616:4.2.1)
<code>clause_title</code>	<code>VARCHAR(200)</code>	nullable	Human-readable clause name
<code>result</code>	<code>ENUM</code>	NOT NULL	<code>Pass</code> , <code>Fail</code> , <code>Pending</code>
<code>remarks</code>	<code>TEXT</code>	nullable	Reviewer remarks
<code>corrective_action</code>	<code>TEXT</code>	nullable	Required corrective action (if Fail)
<code>status</code>	<code>ENUM</code>	NOT NULL	<code>pass</code> , <code>fail</code> , <code>pending</code>
<code>sort_order</code>	<code>SMALLINT</code>	default <code>0</code>	Display ordering
<code>created_at</code>	<code>TIMESTAMPTZ</code>	default <code>now()</code>	

Indexes:

```
CREATE INDEX idx_tr_findings_review_id ON tr_findings(technical_review_id);
CREATE INDEX idx_tr_findings_result ON tr_findings(result);
```

2.15 `tr_queries`

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	PK	Primary key
<code>technical_review_id</code>	<code>UUID</code>	FK → <code>technical_reviews.id</code> , NOT NULL	
<code>question</code>	<code>TEXT</code>	NOT NULL	Query from BIS committee
<code>raised_by</code>	<code>VARCHAR(100)</code>	NOT NULL	Officer/department
<code>raised_at</code>	<code>TIMESTAMPTZ</code>	NOT NULL	
<code>status</code>	<code>ENUM</code>	default <code>'open'</code>	<code>open</code> , <code>resolved</code>
<code>response</code>	<code>TEXT</code>	nullable	
<code>responded_by</code>	<code>UUID</code>	FK → <code>users.id</code> , nullable	
<code>responded_at</code>	<code>TIMESTAMPTZ</code>	nullable	
<code>created_at</code>	<code>TIMESTAMPTZ</code>	default <code>now()</code>	

Indexes:

```
CREATE INDEX idx_tr_queries_review_id ON tr_queries(technical_review_id);
```

2.16 payments

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	
user_id	UUID	FK → users.id, NOT NULL	
payment_no	VARCHAR(30)	UNIQUE, NOT NULL	PAY-YYYY-XXXX
amount	DECIMAL(10,2)	NOT NULL	Payment amount
currency	VARCHAR(3)	default 'INR'	
method	ENUM	nullable	upi , netbanking , card , neft , cash
status	ENUM	default 'pending'	pending , processing , completed , failed , refunded
gateway	VARCHAR(50)	nullable	Razorpay, PayU, etc.
gateway_order_id	VARCHAR(100)	nullable	Gateway's order reference
gateway_payment_id	VARCHAR(100)	nullable	Gateway's payment reference
gateway_response	JSONB	nullable	Raw gateway response
description	VARCHAR(200)	nullable	Payment description
paid_at	TIMESTAMPTZ	nullable	When payment confirmed
refund_amount	DECIMAL(10,2)	nullable	
refunded_at	TIMESTAMPTZ	nullable	
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_payments_application_id ON payments(application_id);
CREATE INDEX idx_payments_user_id ON payments(user_id);
CREATE INDEX idx_payments_status ON payments(status);
```

2.17 notifications

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, NOT NULL	Recipient
type	ENUM	NOT NULL	status_update , document_required , query_raised , payment_due , cert_expiry , govt_query , general
title	VARCHAR(200)	NOT NULL	Notification title
body	TEXT	NOT NULL	Notification body
data	JSONB	nullable	Deep link data (screen, params)
is_read	BOOLEAN	default false	
read_at	TIMESTAMPTZ	nullable	
sent_push	BOOLEAN	default false	Whether push was sent
push_sent_at	TIMESTAMPTZ	nullable	
related_id	UUID	nullable	ID of related entity
related_type	VARCHAR(50)	nullable	application , certificate , payment , etc.
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_notifications_user_id ON notifications(user_id);
CREATE INDEX idx_notifications_unread ON notifications(user_id, is_read) WHERE is_read = false;
CREATE INDEX idx_notifications_created_at ON notifications(created_at DESC);
```

2.18 insights

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
title	VARCHAR(300)	NOT NULL	Article/insight title
slug	VARCHAR(300)	UNIQUE, NOT NULL	URL-friendly slug
summary	TEXT	NOT NULL	AI-generated summary
content	TEXT	nullable	Full article content
category	ENUM	NOT NULL	BIS , EPR , WPC , FSSAI , ISO , CE , general
tags	TEXT[]	default '{}'	Search tags
source	VARCHAR(200)	nullable	Source: "BIS Circular", "MeitY", etc.
source_url	TEXT	nullable	Original source URL
image_url	TEXT	nullable	Cover image (S3)
author	VARCHAR(100)	nullable	Author/AI generated
is_ai_generated	BOOLEAN	default false	AI-summarized circular
is_alert	BOOLEAN	default false	Import/export alert
is_featured	BOOLEAN	default false	Featured in carousel
published_at	TIMESTAMPTZ	NOT NULL	Publication date
read_count	INTEGER	default 0	View counter
bookmark_count	INTEGER	default 0	Bookmark counter
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

Indexes:


```
CREATE INDEX idx_insights_category ON insights(category);
CREATE INDEX idx_insights_published_at ON insights(published_at DESC);
CREATE INDEX idx_insights_is_alert ON insights(is_alert) WHERE is_alert = true;
CREATE INDEX idx_insights_tags ON insights USING GIN(tags);
-- Full-text search
CREATE INDEX idx_insights_fts ON insights USING GIN(to_tsvector('english', title || ' ' || summary));
```

2.19 bookmarks

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, NOT NULL	
insight_id	UUID	FK → insights.id, NOT NULL	
created_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE UNIQUE INDEX idx_bookmarks_user_insight ON bookmarks(user_id, insight_id);
CREATE INDEX idx_bookmarks_user_id ON bookmarks(user_id);
```

2.20 documents (standalone document library)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, NOT NULL	Owner
application_id	UUID	FK → applications.id, nullable	Optional link to app
name	VARCHAR(200)	NOT NULL	Document name
category	VARCHAR(50)	NOT NULL	identity , product , test_report , lab , govt , invoice , other
file_url	TEXT	NOT NULL	S3 URL (presigned for access)
file_key	TEXT	NOT NULL	S3 object key
file_size	INTEGER	NOT NULL	Size in bytes
mime_type	VARCHAR(100)	NOT NULL	
status	ENUM	default 'pending'	pending , approved , rejected
expiry_date	DATE	nullable	For time-limited docs
tags	TEXT[]	default '{}'	
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

Indexes:

```
CREATE INDEX idx_documents_user_id ON documents(user_id);
CREATE INDEX idx_documents_category ON documents(category);
```

2.21 shipments

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PK	Primary key
application_id	UUID	FK → applications.id, NOT NULL	
user_id	UUID	FK → users.id, NOT NULL	
tracking_no	VARCHAR(100)	NOT NULL	Courier tracking number
courier	VARCHAR(100)	NOT NULL	FedEx, Delhivery, etc.
from_address	TEXT	NOT NULL	Origin address
to_address	TEXT	NOT NULL	Destination (lab) address
status	ENUM	default 'booked'	booked , picked_up , in_transit , out_for_delivery , delivered , returned
current_location	VARCHAR(200)	nullable	Last known location
estimated_delivery	DATE	nullable	
delivered_at	TIMESTAMPTZ	nullable	
route	JSONB	nullable	Array of tracking events
created_at	TIMESTAMPTZ	default now()	
updated_at	TIMESTAMPTZ	default now()	

3. Prisma Schema

```

// schema.prisma

generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

// ----- ENUMS -----

enum UserRole {
  admin
  manager
  viewer
}

enum SubscriptionPlan {
  free
  pro
  enterprise
}

enum CertType {
  BIS
  EPR
  WPC
  FSSAI
  ISO
  CE
}

enum ApplicationStatus {
  draft
  submitted
  under_review
  lab_testing
  technical_review
  approved
  rejected
  completed
}

enum DocumentStatus {
  pending
  approved
  rejected
}

enum TRStatus {
  pending
  in_progress
}

```

```

    completed
    failed
}

enum FindingResult {
    Pass
    Fail
    Pending
}

enum PaymentStatus {
    pending
    processing
    completed
    failed
    refunded
}

enum NotificationType {
    status_update
    document_required
    query_raised
    payment_due
    cert_expiry
    govt_query
    general
}

enum InsightCategory {
    BIS
    EPR
    WPC
    FSSAI
    ISO
    CE
    general
}

// ----- MODELS -----

model User {
    id                String    @id @default(uuid())
    phone             String    @unique
    email             String?   @unique
    name              String
    companyName       String    @map("company_name")
    role              UserRole   @default(viewer)
    gstNumber         String?   @unique @map("gst_number")
    panNumber         String?   @map("pan_number")
    address           String?
    city              String?
    state             String?
    pincode           String?
    country            String    @default("India")
    subscription       SubscriptionPlan @default(free)
}

```

```

businessRole      String?      @map("business_role")
industries        String[]     @default([])
targetMarkets     String[]     @default([]) @map("target_markets")
certInterests     String[]     @default([]) @map("cert_interests")
goals            String[]     @default([])
isVerified        Boolean      @default(false) @map("is_verified")
isOnboarded       Boolean      @default(false) @map("is_onboarded")
avatarUrl         String?      @map("avatar_url")
subscriptionExpiresAt DateTime? @map("subscription_expires_at")
createdAt         DateTime      @default(now()) @map("created_at")
updatedAt         DateTime      @updatedAt @map("updated_at")
deletedAt         DateTime?     @map("deleted_at")

// Relations
applications      Application[]
certificates      Certificate[]
notifications     Notification[]
pushTokens        PushToken[]
bookmarks         Bookmark[]
documents         Document[]
payments          Payment[]
refreshTokens     RefreshToken[]

@@map("users")
}

model Application {
  id              String          @id @default(uuid())
  appNo           String          @unique @map("app_no")
  userId          String          @map("user_id")
  certType        CertType        @map("cert_type")
  product         String
  productCategory String?         @map("product_category")
  standard        String
  modelNo         String?         @map("model_no")
  hsnCode         String?         @map("hsn_code")
  status          ApplicationStatus @default(draft)
  progress        Int             @default(0)
  labId           String?         @map("lab_id")
  cbId            String?         @map("cb_id")
  assignedTo      String?         @map("assigned_to")
  assignedOfficer String?         @map("assigned_officer")
  submittedDate   DateTime?       @map("submitted_date")
  estimatedCompletion DateTime?   @map("estimated_completion")
  amount          Decimal         @default(0) @db.Decimal(10, 2)
  paidAmount      Decimal         @default(0) @map("paid_amount") @db.Decimal(10, 2)
  rejectionReason String?         @map("rejection_reason")
  priority        String          @default("normal")
  metadata        Json?
  createdAt       DateTime        @default(now()) @map("created_at")
  updatedAt       DateTime        @updatedAt @map("updated_at")
  deletedAt       DateTime?       @map("deleted_at")

// Relations
user            User              @relation(fields: [userId], references: [id])

```

```

    lab          Lab?          @relation(fields: [labId], references: [id])
    cb           CertificationBody? @relation(fields: [cbId], references: [id])
    timeline     ApplicationTimeline[]
    appDocuments ApplicationDocument[]
    notes        ApplicationNote[]
    queries       ApplicationQuery[]
    payments      Payment[]
    technicalReviews TechnicalReview[]
    certificates Certificate[]
    shipments     Shipment[]

    @@map("applications")
}

model TechnicalReview {
    id          String      @id @default(uuid())
    trNo        String      @unique @map("tr_no")
    applicationId String    @map("application_id")
    status      TRStatus    @default(pending)
    reviewDate  DateTime?   @map("review_date")
    reviewedBy  String?     @map("reviewed_by")
    assignedOfficer String? @map("assigned_officer")
    labReportNo String?     @map("lab_report_no")
    labId       String?     @map("lab_id")
    summary     String?
    passCount   Int         @default(0) @map("pass_count")
    failCount   Int         @default(0) @map("fail_count")
    pendingCount Int        @default(0) @map("pending_count")
    createdAt   DateTime    @default(now()) @map("created_at")
    updatedAt   DateTime    @updatedAt @map("updated_at")

    application Application @relation(fields: [applicationId], references: [id])
    lab          Lab?       @relation(fields: [labId], references: [id])
    findings     TRFinding[]
    queries       TRQuery[]

    @@map("technical_reviews")
}

model TRFinding {
    id          String      @id @default(uuid())
    technicalReviewId String @map("technical_review_id")
    clause      String
    clauseTitle String?     @map("clause_title")
    result      FindingResult
    remarks     String?
    correctiveAction String? @map("corrective_action")
    status      String
    sortOrder   Int         @default(0) @map("sort_order")
    createdAt   DateTime    @default(now()) @map("created_at")

    technicalReview TechnicalReview @relation(fields: [technicalReviewId], references: [id])

    @@map("tr_findings")
}

```



```
model TRQuery {
  id String @id @default(uuid())
  technicalReviewId String @map("technical_review_id")
  question String
  raisedBy String @map("raised_by")
  raisedAt DateTime @map("raised_at")
  status String @default("open")
  response String?
  respondedBy String? @map("responded_by")
  respondedAt DateTime? @map("responded_at")
  createdAt DateTime @default(now()) @map("created_at")

  technicalReview TechnicalReview @relation(fields: [technicalReviewId], references: [id])

  @@map("tr_queries")
}
```

4. Redis Schema

4.1 Key Patterns

KEY PATTERN	TYPE	TTL	DESCRIPTION
session:{userId}	Hash	15 min	Active JWT payload cache
otp:{phone}	String	5 min	OTP value (hashed)
otp:attempts:{phone}	String	5 min	OTP attempt counter
rate:ip:{ip}	String	1 min	IP request count (100/min)
rate:user:{userId}	String	1 min	User request count (1000/min)
cache:insights:list	String	10 min	Cached insights list (JSON)
cache:insights:{id}	String	30 min	Single insight cache
cache:user:{userId}	String	5 min	User profile cache
notif:unread:{userId}	String	—	Unread notification count

4.2 Rate Limiting

Sliding window algorithm:

- IP: 100 requests/minute → 429 Too Many Requests
- Authenticated user: 1,000 requests/minute
- OTP requests: 3 per phone per 5 minutes

5. Database Migrations Strategy

```
migrations/
├─ 001_create_users.sql
├─ 002_create_auth_tables.sql
├─ 003_create_certifications.sql
├─ 004_create_applications.sql
├─ 005_create_application_related.sql
├─ 006_create_technical_reviews.sql
├─ 007_create_payments.sql
├─ 008_create_notifications.sql
├─ 009_create_insights.sql
├─ 010_create_documents.sql
├─ 011_create_shipments.sql
└─ 012_seed_reference_data.sql
```

Migration Rules

1. Always use `IF NOT EXISTS` for CREATE TABLE/INDEX
2. Never drop columns — use soft deprecation + additive columns
3. Every migration is idempotent
4. Test rollback SQL before merging

6. S3 Storage Structure

```
s3://dice-by-sanyog/
├─ avatars/
│   └─ {userId}/{timestamp}.{ext}
├─ documents/
│   └─ {userId}/{applicationId}/{documentId}.{ext}
├─ certificates/
│   └─ {userId}/{certId}/cert.pdf
├─ qr-codes/
│   └─ {certId}/qr.png
└─ insights/
    └─ {insightId}/cover.{ext}
```

Access Pattern: - Upload: Server generates presigned PUT URL (15 min TTL) - Download: Server generates presigned GET URL (60 min TTL) - Public: QR codes and verification images use CloudFront CDN

7. Database Connection Config

```
# .env (server)
DATABASE_URL="postgresql://dice_user:password@localhost:5432/dice_db?schema=public"
REDIS_URL="redis://localhost:6379"
DATABASE_POOL_MIN=2
DATABASE_POOL_MAX=10
DATABASE_CONNECTION_TIMEOUT=5000
```

```
// Prisma connection
const prisma = new PrismaClient({
  log: ['query', 'error', 'warn'],
  datasources: {
    db: { url: process.env.DATABASE_URL }
  }
});
```